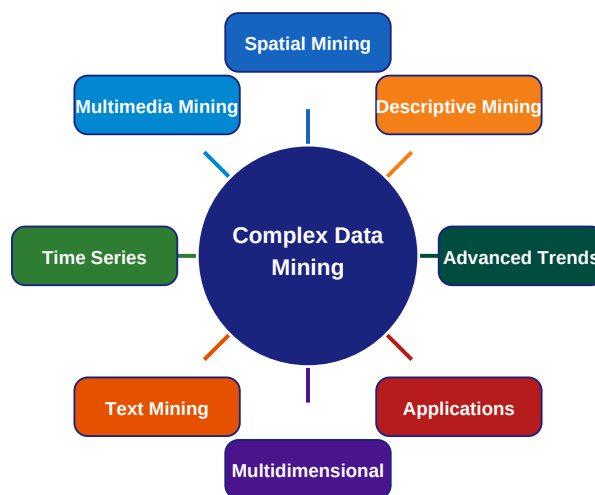


COMPLEX DATA MINING

Comprehensive Study Notes

8 Modules · 16 Lectures · All Topics Covered



Covers: Mining Complex Types of Data | Applications & Trends | Spatial, Multimedia, Time Series, Text Mining

TABLE OF CONTENTS

Module 1	Introduction to Complex Data (Lectures 1–2)	Pg 3
Module 2	Descriptive Mining of Complex Objects (Lectures 3–4)	Pg 7
Module 3	Spatial Data Mining (Lectures 5–6)	Pg 11
Module 4	Multimedia Data Mining (Lectures 7–8)	Pg 15
Module 5	Time Series & Sequence Mining (Lectures 9–10)	Pg 19
Module 6	Text Mining (Lectures 11–12)	Pg 23
Module 7	Applications of Data Mining (Lectures 13–14)	Pg 27
Module 8	Advanced Themes & Trends (Lectures 15–16)	Pg 31

MODULE 1: Introduction to Complex Data (Lectures 1–2)

Lecture 1: What is Complex Data?

1.1 Definition and Overview

Data Mining is the process of extracting novel, valid, useful, and understandable patterns from large volumes of data. Traditional data mining focused on structured, tabular data — rows and columns in relational databases. **Complex Data** refers to data that does not fit neatly into relational tables and requires specialized techniques for storage, processing, indexing, and pattern discovery.

Definition: Complex data is any data that deviates from simple, flat relational tables due to its structure, semantics, or volume. This includes spatial coordinates, images, audio waveforms, time-ordered measurements, and natural language text.

1.2 Traditional vs Complex Data

Aspect	Comparison
Traditional Data	Complex Data
Structured, tabular format	Semi-structured or unstructured
Fixed schema (rows & columns)	Variable or no schema
Simple data types (int, float, string)	Rich data types (images, audio, text)
Easy to index with B-trees	Requires specialized indexing (R-tree, inverted index)
SQL queries sufficient	Requires advanced algorithms
Low dimensionality	High dimensionality (hundreds to thousands of features)
Examples: Sales records, HR data	Examples: Satellite images, social media posts

1.3 Types of Complex Data

Modern data sources generate diverse and complex data types. The four primary categories studied in this course are:

Type	Description	Example
Spatial Data	Data with geographic coordinates or location attributes. Includes maps, GPS traces, satellite imagery. Stored in spatial databases. Requires spatial indexing.	GIS maps, urban sensor data
Multimedia Data	Images, audio clips, and video recordings. Rich in content but unstructured. Feature extraction is essential before mining.	Medical scans, surveillance video, music libraries
Time-Series Data	Sequence of measurements recorded over time. Characterized by trend, seasonality, and noise. Temporal ordering matters.	Stock prices, ECG signals, weather readings

Text Data	Natural language documents, web pages, social media. Highly unstructured. Requires NLP preprocessing before analysis.	News articles, emails, tweets
-----------	---	-------------------------------

1.4 Challenges in Complex Data Mining

High Dimensionality: Complex data often has hundreds to thousands of features (dimensions). The 'curse of dimensionality' causes distance metrics to become meaningless and algorithms to slow dramatically. Feature selection and dimensionality reduction (PCA, LDA) are critical.

Heterogeneity: Complex data comes from multiple sources with different formats, scales, and semantics. Integrating spatial, text, and numerical data requires common representations and schema matching.

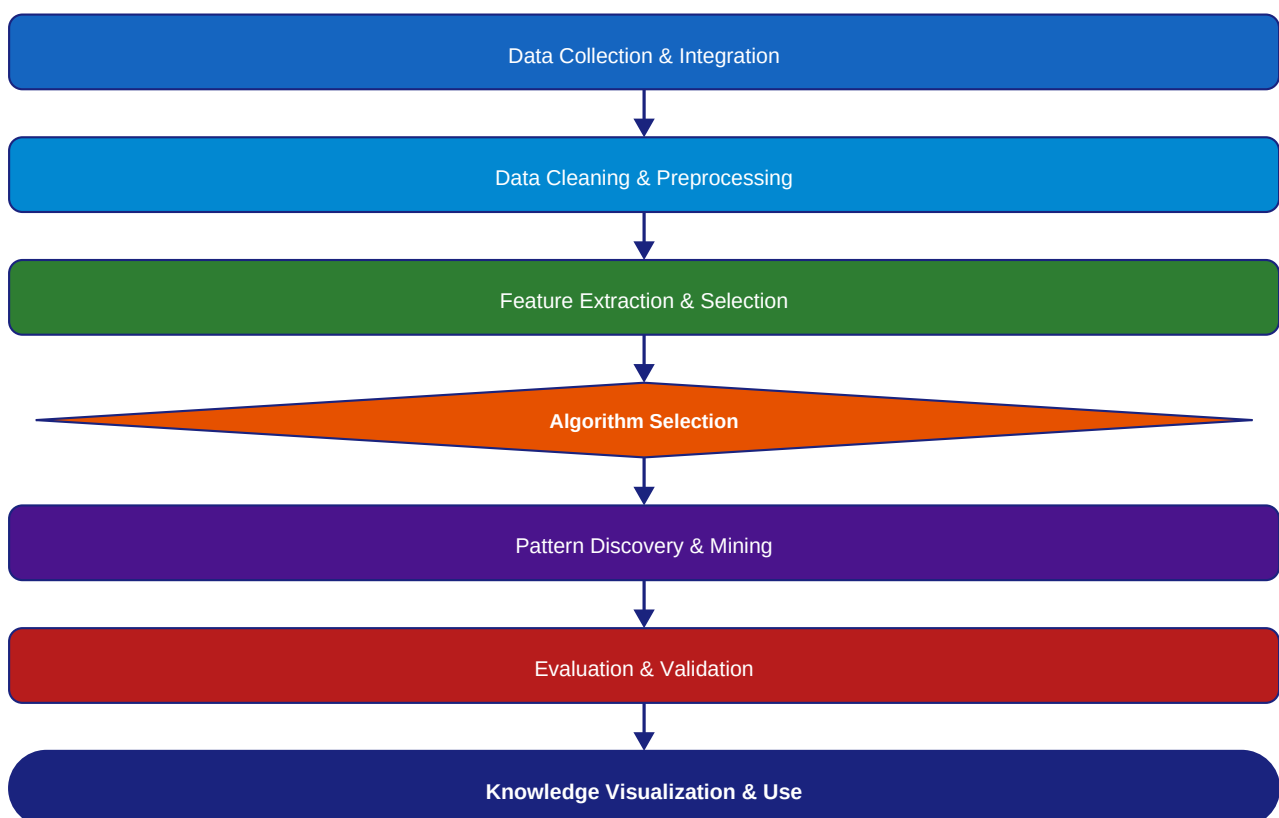
Scalability: Datasets grow to terabytes/petabytes. Algorithms must scale with $O(n \log n)$ or better. Distributed computing (Hadoop, Spark) and approximate methods become necessary.

Noise & Incompleteness: Real-world complex data is noisy (sensor errors, OCR mistakes) and often incomplete. Robust mining algorithms must tolerate uncertainty.

Semantic Gap: There is often a gap between low-level features (pixel values) and high-level concepts (a face, a sentiment). Bridging this gap requires deep feature engineering or deep learning.

1.5 Flowchart: General Data Mining Process

General Data Mining Workflow



Lecture 2: Multidimensional Analysis

2.1 The Data Cube Concept

A **Data Cube** is a multidimensional data model that organizes data by dimensions (like Time, Location, Product) and measures (like Sales, Revenue). It extends the two-dimensional spreadsheet into n-dimensional space, enabling fast, flexible aggregation and analysis across multiple dimensions simultaneously.

A data cube with 3 dimensions (Time × Location × Product) and the measure Sales forms a 3D cube. Each cell in the cube holds an aggregated value such as total sales for a specific time, location, and product combination.

2.2 Key Terminology

Term	Definition
Dimension	An attribute used to categorize data (e.g., Time, Region, Product)
Measure	A numerical fact being analyzed (e.g., Sales, Profit, Count)
Cell	The intersection of dimension values holding a measure
Cuboid	A group of cells at a specific aggregation level; the full cube has 2^n cuboids for n dimensions
Lattice	Hierarchical structure showing all possible cuboids from finest to coarsest granularity
Apex	The topmost cuboid — contains a single grand total across all dimensions
Base Cuboid	The lowest level cuboid with all dimension details

2.3 OLAP Operations in Detail

OLAP (Online Analytical Processing) provides a set of operations to navigate and explore the data cube interactively:

Roll-Up (Drill-Up)

Aggregates data by climbing a dimension hierarchy or reducing a dimension. Example: Summarizing daily sales data into monthly or yearly totals. Mathematically, it applies an aggregate function (SUM, AVG, COUNT) at a coarser granularity. Roll-up reduces the number of dimensions visible in the analysis.

Example: Daily → Monthly → Yearly; City → State → Country

Drill-Down

The reverse of roll-up — navigates from a summarized view to a more detailed view by descending a hierarchy or introducing a new dimension. Enables pinpointing the cause of anomalies found at the aggregate level. Useful in root cause analysis.

Example: Year → Quarter → Month → Day

Slice

Selects a single value from one dimension, producing a sub-cube with one fewer dimension. Equivalent to a WHERE clause on one dimension. Reduces the cube to a planar view (a 'slice').

Example: Slice on Time = 'Q1 2024' → 2D matrix of Product × Region

Dice

Defines a sub-cube by selecting ranges on two or more dimensions simultaneously. Equivalent to multiple WHERE clauses. The resulting sub-cube retains the same number of dimensions but with restricted ranges.

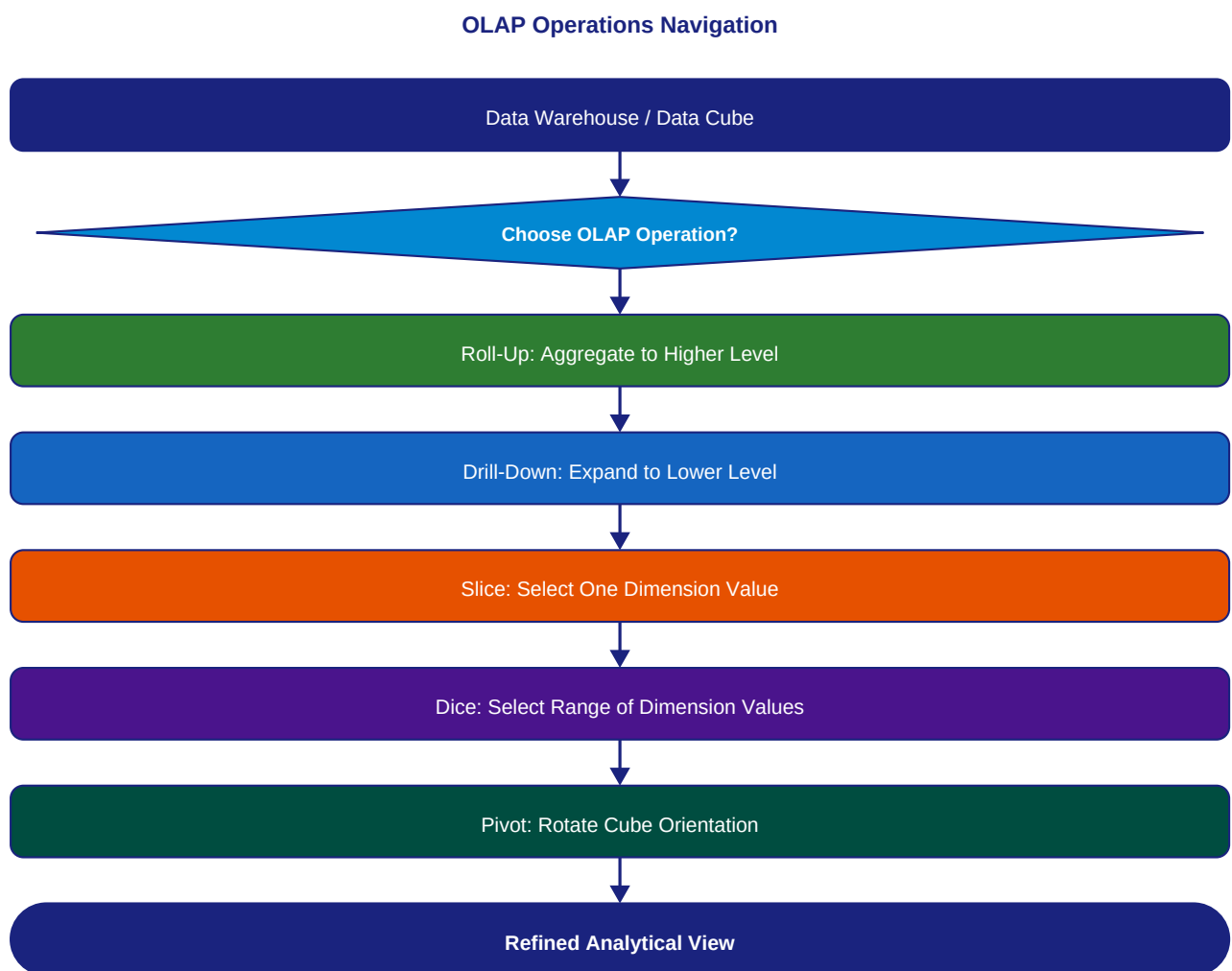
Example: Dice: Time ∈ {Q1, Q2}, Region ∈ {North, South}

Pivot (Rotate)

Rotates the data cube to provide an alternative presentation by changing the orientation of the axes. Equivalent to transposing a matrix. Useful for comparing different perspectives of the same data.

Example: Swap rows and columns in a cross-tabulation report

2.4 OLAP Operations Flowchart



2.5 Applications of Multidimensional Analysis

Key Application Areas of Multidimensional Analysis



Retail: Retailers use OLAP cubes to analyze sales by store, product category, and time period. Roll-up from individual transactions to daily totals, then monthly, then quarterly for executive reports. Drill-down reveals which specific SKUs drove revenue increases in a particular region.

Finance: Banks maintain cubes of transaction data across branches, account types, and time periods. OLAP enables rapid computation of risk-weighted assets, non-performing loan ratios, and profitability metrics at different levels of granularity.

MODULE 2: Descriptive Mining of Complex Objects (Lectures 3–4)

Lecture 3: Descriptive Mining Concepts

3.1 What is Descriptive Mining?

Descriptive mining (also called descriptive data mining) discovers general properties and summaries of a dataset rather than predicting outcomes. It provides a compact, human-interpretable description of the data, helping analysts understand the overall distribution, class characteristics, and key differences between groups.

Goal: Characterize data as a whole or distinguish one class from another using compact summaries, generalized descriptions, and concept hierarchies.

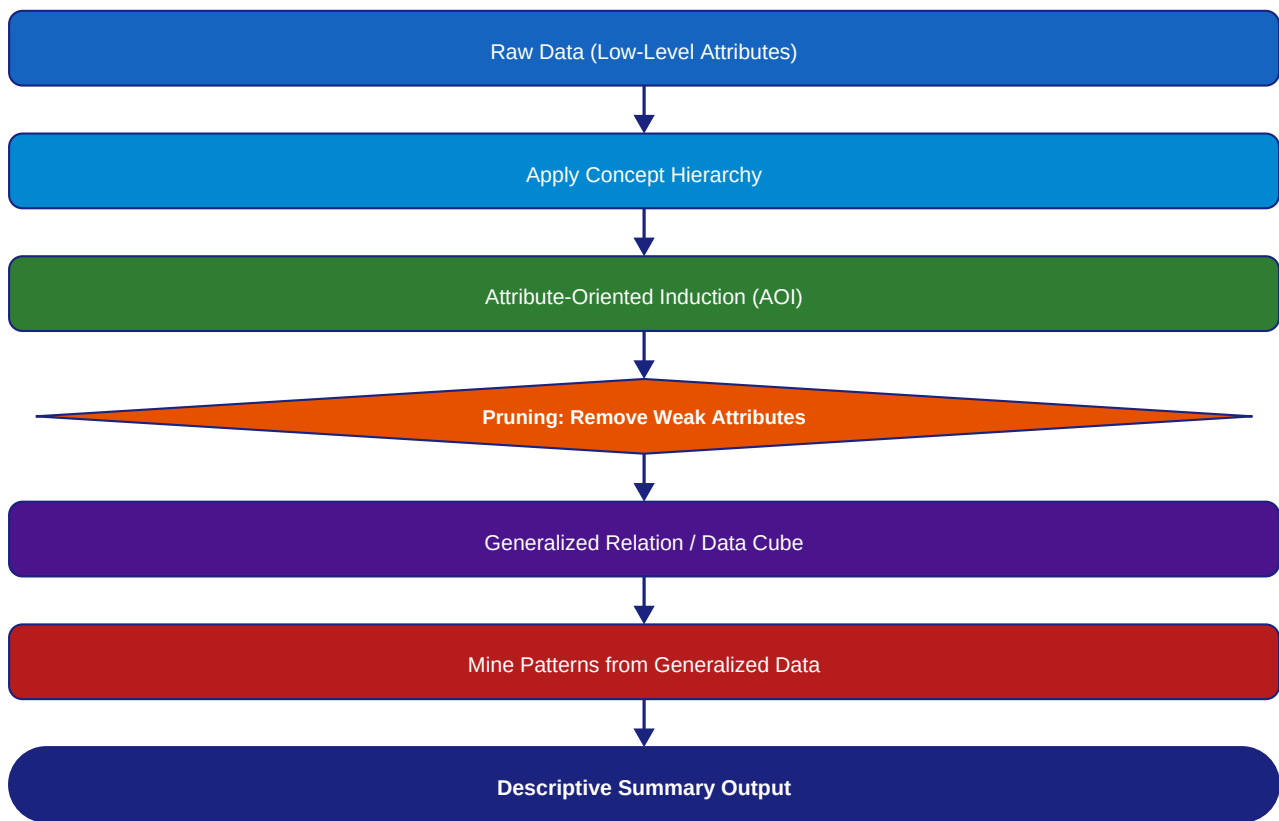
3.2 Characterization vs Discrimination

Characterization	Discrimination
Data Characterization	Data Discrimination
Summarizes target class data	Compares target vs. contrast classes
Answers: 'What is this class like?'	Answers: 'How does class A differ from B?'
Produces generalized descriptions	Produces discriminant rules
Uses concept hierarchy generalization	Uses comparative attribute weighting
Output: Class description table	Output: Discriminant predicates
Example: Profile of 'high-value customers'	Example: High-value vs low-value customers

3.3 Data Generalization Techniques

Generalization replaces low-level, specific data values with higher-level, more abstract concepts using concept hierarchies. This is fundamental to descriptive mining because it allows concise summaries.

Data Generalization via AOI



Attribute-Oriented Induction (AOI): Starting with a relational query result, AOI repeatedly replaces specific attribute values with more general ones using concept hierarchies. For example, age values 22, 23, 24 → 'Young'; cities 'Delhi', 'Mumbai' → 'India'. The process continues until the relation is compact enough to form a summary table.

Example Concept Hierarchy for Location: Specific Address → City → State → Country → Continent → World

3.4 Summarization Methods

Method	Description
Statistical Summary	Mean, median, mode, variance, standard deviation, quartiles. Captures central tendency and spread.
Concept Description	Generates a linguistic description of a class using AOI and concept hierarchies.
Association Rules	Discovers co-occurrence patterns: 'If buys X, then also buys Y with confidence c%.
Histograms & Distributions	Visual summaries showing frequency of values across attribute ranges.
Decision Tree Summaries	Compact tree structures that describe class membership conditions.

Lecture 4: Mining Complex Data Objects

4.1 Structured vs Semi-Structured vs Unstructured Data

Type	Characteristics
Structured	Fixed schema in relational tables. SQL queryable. Easy to mine with classical algorithms. Examples: Employee database, sensor logs in relational DB.
Semi-Structured	Has some structure but flexible schema. Examples: XML, JSON, HTML, email headers. Mining requires parsing and schema inference.
Unstructured	No predefined schema. Includes free text, images, audio, video. Mining requires specialized feature extraction before pattern discovery.

4.2 Feature Extraction Techniques

Feature extraction transforms raw complex data into a numerical feature vector that standard mining algorithms can process. This is the most critical step in mining complex data.

Feature Extraction Pipeline



Data Type	Common Features Extracted
Image Features	Color histograms, edge detection, texture descriptors (GLCM, Gabor filters), shape moments, SIFT/SURF keypoints
Audio Features	Mel-frequency cepstral coefficients (MFCCs), zero-crossing rate, spectral centroid, chroma features
Text Features	Bag of Words (BoW), TF-IDF vectors, word embeddings (Word2Vec, GloVe), n-gram counts
Spatial Features	Coordinates, bounding boxes, area, perimeter, convex hull, spatial adjacency
Time-Series Features	Statistical moments, autocorrelation, FFT coefficients, wavelet coefficients, DTW distance

4.3 Similarity Measures

Mining complex objects requires quantifying similarity or distance between objects. Different data types require different similarity measures:

Measure	Formula / Method	Best Used For
Euclidean Distance	$d(x,y) = \sqrt{\sum (x_i - y_i)^2}$	Continuous numerical data, feature vectors

Cosine Similarity	$\cos(\theta) = (x \cdot y) / (x y)$	Text documents, high-dimensional sparse vectors
Jaccard Similarity	$J(A,B) = A \cap B / A \cup B $	Sets, binary features
Edit Distance (Levenshtein)	Min edits to transform $s1 \rightarrow s2$	Strings, DNA sequences
DTW Distance	Dynamic Time Warping	Time series of different lengths
Hausdorff Distance	Max of min distances between point sets	Shapes, spatial objects
Earth Mover's Distance	Optimal transport cost between distributions	Image histograms, distributions

✓ Choosing the right similarity measure is crucial: using Euclidean distance on text vectors can produce poor clustering results compared to cosine similarity.

MODULE 3: Spatial Data Mining (Lectures 5–6)

Lecture 5: Introduction to Spatial Databases

5.1 What is Spatial Data?

Spatial data represents objects or phenomena that have a position in space, described by geometric coordinates. Unlike tabular data, spatial objects possess **location**, **geometry**, and **spatial relationships** with other objects. Spatial databases extend relational databases with spatial data types and operators.

Spatial data is any data that describes objects in terms of their location in a coordinate reference system (CRS), typically geographic (latitude/longitude) or projected (x/y Cartesian coordinates).

5.2 Spatial Data Types

Type	Description	Example
Points	Zero-dimensional objects representing a single location. Described by (x, y) or (lat, lon). Examples: ATM locations, fire stations, earthquake epicenters, IP geolocations.	(77.2090, 28.6139) — New Delhi
Lines / Polylines	One-dimensional objects representing paths or boundaries formed by sequences of points. Examples: Roads, rivers, pipelines, flight paths, coastlines.	Road from point A to B through waypoints
Polygons	Two-dimensional closed regions representing areas. Defined by a closed set of points. Examples: Country borders, forest coverage, flood zones, building footprints.	State of Rajasthan boundary polygon
Rasters	Grid-based representation where space is divided into uniform cells (pixels), each holding a value. Examples: Satellite images, digital elevation models (DEMs), land-cover maps.	30m resolution SRTM elevation raster
3D Objects	Objects with depth in addition to x and y. Used in urban modeling, subsurface geology, and indoor mapping. Examples: Building 3D models (CityGML), mine shafts.	3D building model of a skyscraper

5.3 Spatial Relationships

Spatial relationships describe how objects relate to each other in space. These are fundamental for spatial queries and mining:

Distance-based

Measures proximity between objects. Euclidean distance for flat surfaces; Great-circle (Haversine) distance for Earth's curvature. Used in queries like 'Find all hospitals within 5 km of the accident site.'

Formula: $d = 2R \cdot \arcsin(\sqrt{[\sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2)]})$

Direction-based

Describes the angular or compass direction from one object to another. Cardinal directions (N, S, E, W) or angular bearings (0°–360°). Used in navigation and spatial reasoning.

Formula: Bearing = $\text{atan2}(\sin(\Delta\lambda) \cdot \cos(\phi_2), \cos(\phi_1) \cdot \sin(\phi_2) - \sin(\phi_1) \cdot \cos(\phi_2) \cdot \cos(\Delta\lambda))$

Topological

Describes structural relationships that are invariant under continuous deformation. The DE-9IM model defines relationships: *Disjoint*, *Meet*, *Overlap*, *Covers*, *CoveredBy*, *Contains*, *Within*, *Equals*. Essential for GIS analysis.

Formula: Forest polygon Contains wildlife reserve polygon

5.4 Spatial Indexing Structures

Efficient spatial queries require specialized index structures:

Index Structure	Description
R-Tree	Hierarchical bounding rectangle index. Most widely used for spatial databases. Supports range and nearest-neighbor queries.
R*-Tree	Improved R-Tree with better node packing and split strategies. Reduces overlap between bounding boxes.
Quad-Tree	Recursive space partitioning into quadrants. Simple and effective for point data. Can be unbalanced for skewed distributions.
KD-Tree	Binary space partitioning tree for k-dimensional data. Very efficient for k-NN queries on point clouds.
Grid Index	Uniform grid overlay. Simple and fast for uniformly distributed data. Wastes space for skewed distributions.

Lecture 6: Spatial Data Mining Techniques

6.1 Spatial Clustering

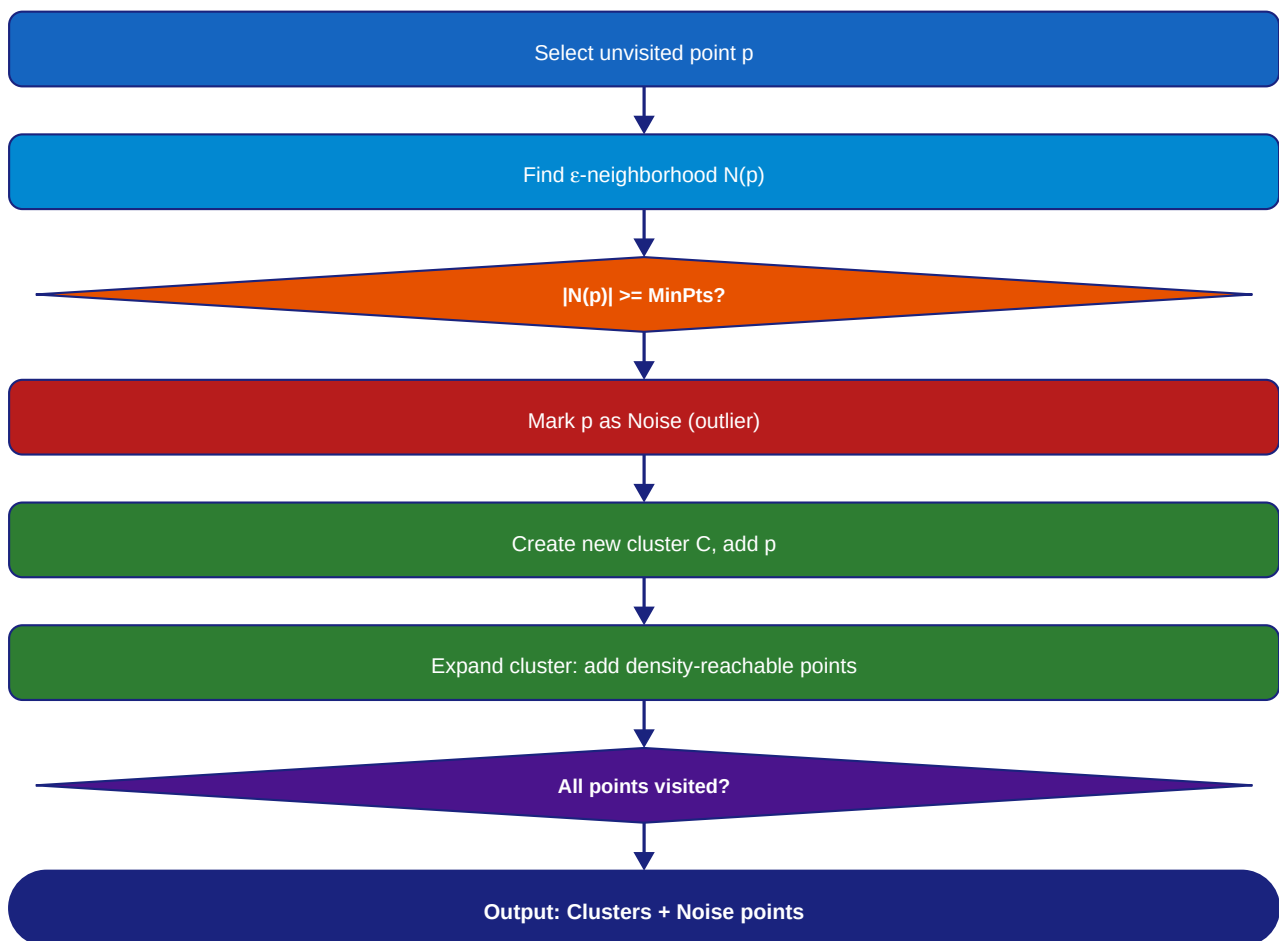
Spatial clustering groups nearby geographic objects into clusters. Unlike traditional clustering, spatial clusters must account for geographic proximity and can have irregular shapes.

Algorithm	Description
DBSCAN	Density-Based Spatial Clustering of Applications with Noise. Groups points within ϵ distance with at least MinPts neighbors. Discovers arbitrary-shaped clusters and labels outliers as noise. Very popular for geographic data.
CLARANS	Clustering Large Applications based on Randomized Search. Extends k-medoids with random sampling. Designed for large spatial datasets. Considers spatial locality.

STING	STatistical INformation Grid. Divides space into rectangular cells in a hierarchy. Mines statistical information stored in each cell. Very fast for large datasets.
WaveCluster	Uses wavelet transforms to find dense regions at multiple granularities. Handles noise well and is efficient for large datasets.

6.2 DBSCAN Algorithm Flowchart

DBSCAN Spatial Clustering Algorithm



6.3 Spatial Classification

Spatial classification assigns geographic objects to predefined classes based on both their non-spatial attributes AND their spatial context (neighboring objects, location-based features). Traditional classifiers (decision trees, SVM, neural nets) are extended to incorporate spatial autocorrelation — the tendency of nearby objects to have similar properties (Tobler's First Law of Geography).

- Spatial Decision Trees: Splits based on spatial predicates (e.g., 'is within flood zone')
- Spatial Autoregressive Models: Account for spatial dependency in class labels
- Random Forest with Spatial Features: Adds lat/lon, distance to landmarks as features
- Geographically Weighted Regression (GWR): Local regression model where coefficients vary across space

6.4 Spatial Association Rules

Spatial association rules discover interesting co-location patterns among spatial objects. Unlike traditional association rules on transactions, spatial rules express geographic proximity as the 'basket'.

Example Spatial Association Rule: $is_a(X, Hospital) \wedge close_to(X, Y, 2km) \rightarrow is_a(Y, Pharmacy)$ [support: 65%, confidence: 78%] — Meaning: 78% of locations close to hospitals also have a pharmacy within 2 km.

Co-location Patterns: A co-location pattern is a set of feature types whose instances are frequently located in close proximity. The Co-location Miner algorithm uses participation ratios instead of support to measure prevalence.

6.5 Applications of Spatial Data Mining

Spatial Mining Application Domains



Urban Planning: Mining spatial patterns in traffic flow, population density, land use, and infrastructure helps city planners optimize public transport routes, identify areas needing new schools or hospitals, and predict urban growth patterns.

Epidemiology: Identifying spatial clusters of disease outbreaks (e.g., cholera cases clustered around contaminated water sources) enables rapid public health response. John Snow's 1854 cholera map was an early example of spatial data mining.

MODULE 4: Multimedia Data Mining (Lectures 7–8)

Lecture 7: Multimedia Databases

7.1 Overview of Multimedia Data

Multimedia data encompasses digital content in multiple formats: images, audio, video, and animations. Unlike text or numerical data, multimedia content is primarily perceived by human senses (sight, hearing) and carries rich semantic meaning that is difficult to quantify automatically — this is the **semantic gap**.

The Semantic Gap: The difference between low-level features that a computer can compute (pixel RGB values, audio waveform amplitudes) and high-level semantic concepts that humans perceive (a smiling face, a cheerful melody, a violent scene).

7.2 Types of Multimedia Data

Type	Description	Examples
Images	Static 2D visual data. Represented as pixel matrices (raster images) or mathematical equations (vector images). RGB color model most common. File formats: JPEG, PNG, GIF, BMP, TIFF.	Medical X-rays, satellite photos, product images, artwork
Audio	Time-varying acoustic signals. Represented as waveforms (PCM) or compressed formats. Key attributes: sampling rate (Hz), bit depth, channels (mono/stereo). Formats: WAV, MP3, FLAC, AAC.	Music recordings, voice calls, environmental sound monitoring
Video	Sequence of image frames with synchronized audio. Characterized by resolution, frame rate (fps), duration, and codec. Very large data volumes (HD video: ~1.5 GB/min uncompressed). Formats: MP4, AVI, MKV, MOV.	CCTV surveillance, movies, medical endoscopy videos
3D & Animation	Three-dimensional models represented as meshes, point clouds, or voxels. Animated sequences add temporal dimension to 3D. Formats: OBJ, STL, FBX.	Medical 3D CT/MRI scans, game character models

7.3 Challenges: Storage, Indexing, and Retrieval

Storage Challenges:

- Volume: A single uncompressed 4K video at 30fps generates ~6 GB/minute. Large-scale collections require distributed storage (HDFS, object stores like Amazon S3).
- Compression: Lossy (JPEG, MP3) vs. lossless (PNG, FLAC) trade-offs between storage efficiency and data quality.
- Metadata management: Storing descriptive metadata (EXIF for images, ID3 tags for audio) alongside content for organization.

Indexing Challenges:

- Feature-based indexing: Must first extract features, then index the feature vectors (using techniques like LSH, KD-trees, or VP-trees).
- Multi-modal indexing: Different media types require different index structures.
- Dynamic updates: Multimedia databases grow continuously; indices must support efficient insertions.

Retrieval Challenges:

- Content-Based Image Retrieval (CBIR): Retrieve images based on visual similarity rather than keywords. Query by example.
- Audio Fingerprinting: Identify a song from a short snippet (as used by Shazam).
- Video Temporal Search: Find specific scenes or events within long video recordings.

Lecture 8: Multimedia Mining Techniques

8.1 Image Feature Extraction

Image features encode visual information in numerical vectors. Three main categories of low-level image features are:

Color Features

Color Histograms: Count pixel frequencies in each color bin. Invariant to scale and rotation. Fast to compute. **Color Moments:** Mean, variance, skewness of color distribution per channel. Compact representation. **Dominant Colors:** K-means on pixel colors to find representative palette. **Color Correlograms:** Joint probability of colors at specific spatial distances.

Texture Features

GLCM (Gray-Level Co-occurrence Matrix): Encodes spatial relationships between pixel intensity pairs. Haralick features: contrast, correlation, energy, homogeneity derived from GLCM. **Gabor Filters:** Frequency and orientation-sensitive filters mimicking human visual cortex. **LBP (Local Binary Patterns):** Encode local texture by comparing pixels to their neighbors.

Shape Features

Contour-based: Fourier descriptors, chain codes describe object boundaries. **Region-based:** Geometric moments (area, centroid, eccentricity), Zernike moments. **SIFT / SURF:** Scale-Invariant Feature Transform — detects and describes local keypoints invariant to scale, rotation, and illumination changes. Widely used in image matching.

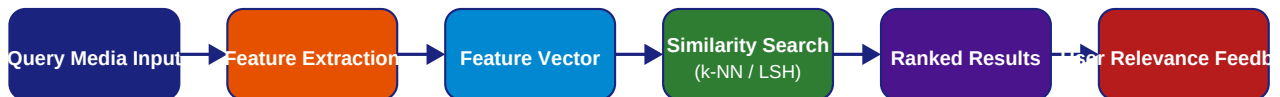
8.2 Audio Feature Extraction

Audio Feature	Description & Use
MFCC (Mel-Frequency Cepstral Coefficients)	Mimic human auditory perception. 13–40 coefficients capture timbral characteristics. Standard feature for speech and music classification.
Zero-Crossing Rate (ZCR)	Rate at which the signal changes sign. High ZCR → percussive sounds. Useful for distinguishing voiced/unvoiced speech.
Spectral Centroid	Weighted mean of frequencies present. Indicates 'brightness' of sound. Higher centroid → brighter sound.
Spectral Roll-off	Frequency below which X% (usually 85%) of spectral energy lies. Distinguishes harmonic from percussive content.

Chroma Features	12-dimensional vector representing energy in each pitch class (C, C#, D...). Useful for chord recognition and music harmonics analysis.
Beat & Tempo	BPM (beats per minute) estimation using onset detection. Key feature for music retrieval and genre classification.

8.3 Similarity Search & Multimedia Retrieval Pipeline

Content-Based Multimedia Retrieval Pipeline



Locality-Sensitive Hashing (LSH): An approximate nearest-neighbor technique that hashes similar high-dimensional vectors to the same bucket with high probability. Dramatically reduces search time from $O(n)$ to $O(1)$ on average for large multimedia databases.

Relevance Feedback (RF): An interactive retrieval refinement technique where the user marks retrieved results as relevant or irrelevant, and the system adjusts the query vector accordingly (e.g., Rocchio algorithm). This bridges the semantic gap over multiple interaction rounds.

8.4 Applications

Application	Technical Description
Face Recognition	Extract facial features (Eigenfaces, LBP, deep CNN embeddings), match against database. Applications: security systems, phone unlock, law enforcement.
Video Surveillance	Detect moving objects, classify behaviors (running, falling), identify persons of interest, track trajectories across cameras.
Medical Image Analysis	Classify tumors in MRI/CT scans, detect anomalies in X-rays, segment organs. CAD (Computer-Aided Diagnosis) systems.
Music Information Retrieval	Query by humming, automatic genre classification, mood detection, playlist generation based on audio similarity.
Video Summarization	Automatically identify key frames/scenes from long videos. Shot boundary detection, highlight extraction for sports.

MODULE 5: Time Series & Sequence Mining (Lectures 9–10)

Lecture 9: Time Series Data

9.1 Definition and Characteristics

A **time series** is an ordered sequence of real-valued measurements collected at uniform or non-uniform time intervals. The critical property distinguishing time series from ordinary numerical data is that the **temporal order carries information** — shuffling the observations destroys meaningful patterns.

Formal Definition: A time series $T = \{t_1, t_2, t_3, \dots, t_n\}$ where each t_i is a real number recorded at time stamp i . The series may be univariate (one variable) or multivariate (multiple variables measured simultaneously).

9.2 Components of Time Series

Trend

Long-term systematic increase or decrease in the series over time. Captured by fitting a linear, polynomial, or exponential function to the data. Example: Gradual increase in global mean temperature over decades.

Seasonality

Regular, predictable fluctuations that repeat at known fixed periods (hourly, daily, weekly, annually). Example: Higher ice cream sales every summer. Removed using seasonal decomposition (STL, X-11/X-13-ARIMA-SEATS).

Cyclic Patterns

Irregular fluctuations not tied to a fixed period. Usually related to economic or business cycles with varying duration. Example: Economic recessions occurring every 5–10 years. Harder to model than seasonality.

Irregular/Noise

Random, unpredictable short-term variation remaining after removing trend, seasonality, and cycles. Caused by measurement error, random external events. Cannot be predicted — best modeled stochastically.

9.3 Time Series Preprocessing

Before applying mining algorithms, time series often require preprocessing:

Technique	Description
Normalization	Scale values to $[0,1]$ or zero-mean unit-variance to enable comparison across series with different scales.
Resampling	Convert series to uniform time intervals. Upsampling (interpolation) or downsampling (aggregation).
Noise Removal	Apply smoothing filters: Moving average, Gaussian smoothing, or Savitzky-Golay filter to remove high-frequency noise.

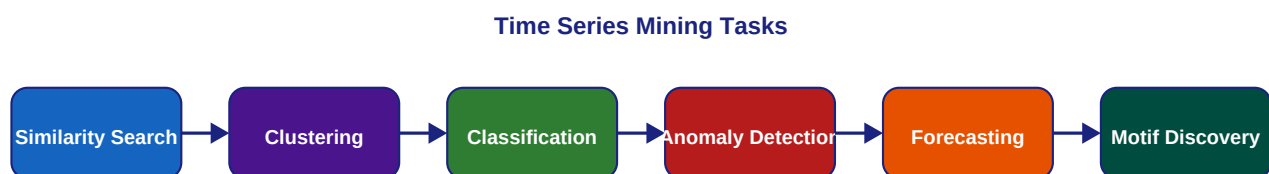
Missing Value Imputation	Fill gaps using linear interpolation, spline interpolation, or model-based imputation (LOCF, NOCB).
Dimensionality Reduction	PAA (Piecewise Aggregate Approximation), SAX (Symbolic Aggregate approXimation), or DWT (Discrete Wavelet Transform).

9.4 Similarity Measures for Time Series

Comparing time series requires specialized distance measures that handle varying lengths, time shifts, and warping:

Measure	Description	Formula
Euclidean Distance	Point-by-point squared difference sum. Fast ($O(n)$) but sensitive to noise, shifts, and scaling. Requires equal-length series aligned in time.	$d(Q, C) = \sqrt{\sum (q_i - c_i)^2}$
Dynamic Time Warping (DTW)	Allows elastic alignment — matches each point in Q to the best-aligned point in C using dynamic programming. Handles time shifts and speed variations. More robust than Euclidean for most real-world time series. $O(n^2)$ complexity.	$DTW(Q, C) = \min \text{warping path cost}$
Longest Common Subsequence (LCSS)	Finds longest subsequence common to both series (with tolerance ϵ). Very robust to noise and outliers. Does not require equal-length series.	$LCSS(Q, C, \epsilon) = \text{longest matching subsequence}$
Fréchet Distance	Measures similarity of curves considering ordering. Intuitively: minimum leash length needed to walk a dog along both curves simultaneously.	$\min \text{ over all traversals of } \max d(\alpha(t), \beta(t))$

9.5 Time Series Mining Tasks



Anomaly/Outlier Detection: Finding unusual subsequences or time points. Matrix Profile (STOMP algorithm) efficiently finds all pairwise distances to identify motifs (recurring patterns) and discords (unusual patterns). Critical for fraud detection in financial transactions, predictive maintenance in manufacturing, and network intrusion detection.

Lecture 10: Sequence Mining

10.1 Sequential Pattern Mining

Sequential pattern mining discovers frequently occurring ordered event sequences in a sequence database. Unlike time series (continuous numeric values), sequences consist of discrete events/items ordered by time. The goal is to find patterns that appear in at least minSup fraction of sequences.

Example: In a customer purchase database, find that customers who buy Laptop → Antivirus → Printer cover occur frequently (support > 40%). This is a sequential pattern of length 3.

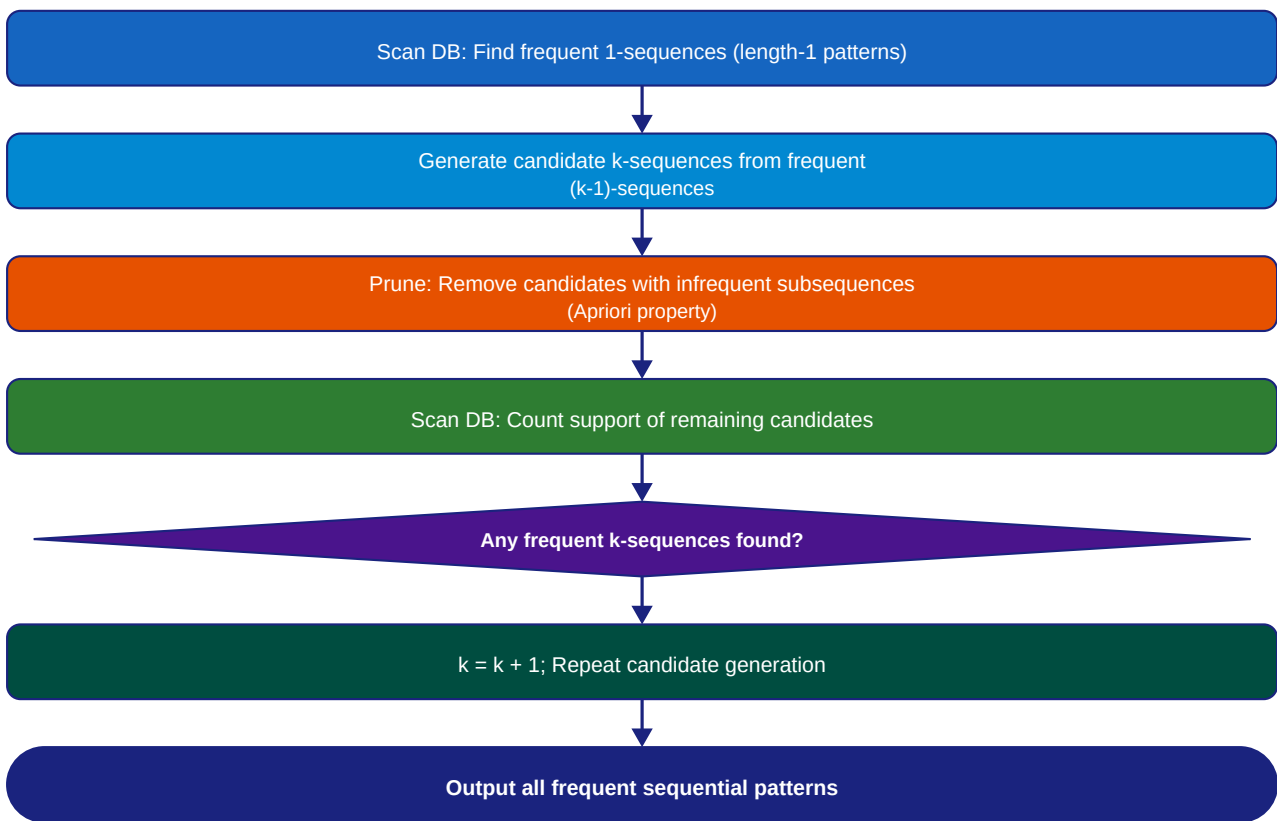
10.2 Key Definitions

Term	Definition
Sequence	An ordered list of events/itemsets. E.g., ■(AB)(C)(DE)■
Subsequence	s1 is a subsequence of s2 if all elements of s1 appear in s2 in order (not necessarily contiguous)
Support	Fraction of sequences in the database that contain a given pattern
Sequential Pattern	A subsequence with support $\geq$ minSup threshold
maximal Pattern	A sequential pattern not subsumed by a longer frequent pattern
closed Pattern	A sequential pattern whose proper subsequences all have lower support

10.3 Algorithm 1: GSP (Generalized Sequential Patterns) — Apriori-based

GSP is an Apriori-based algorithm that uses a candidate generation-and-test approach:

GSP Algorithm Flowchart

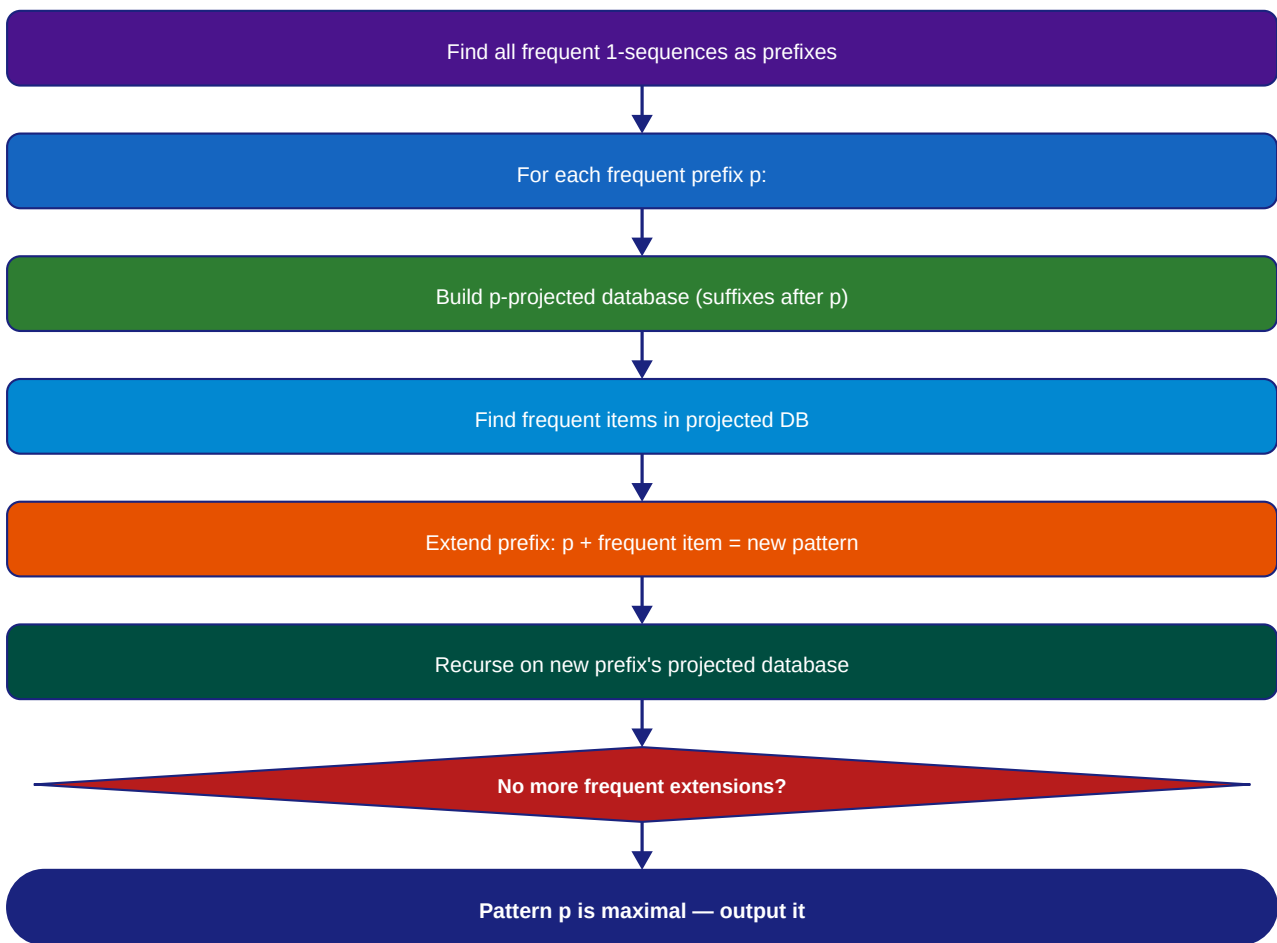


Limitations of GSP: Multiple database scans (one per pattern length). Generates a huge number of candidates. Poor performance when patterns are long or minSup is very low.

10.4 Algorithm 2: PrefixSpan (Pattern-Growth)

PrefixSpan avoids candidate generation entirely. It mines patterns by prefix extension and projects the database into smaller projected databases.

PrefixSpan Algorithm Flowchart



Advantages of PrefixSpan: No candidate generation. Only 2 database scans (one full scan + projected scans). Much more memory and time efficient than GSP for long patterns. One of the most widely used sequence mining algorithms.

10.5 Applications of Sequence Mining

Application	Description
Stock Market Prediction	Mine historical price movement sequences to predict future trends. E.g., 'Three consecutive closes above 20-day moving average precede rally.'
User Behavior Analysis	Analyze web click-stream sequences to understand navigation patterns, detect fraud (anomalous transaction sequences), personalize recommendations.
DNA Sequence Analysis	Find frequent motifs in genomic sequences that correspond to regulatory elements or protein-coding regions.
Medical Event Sequences	Analyze sequences of diagnoses, treatments, and outcomes to discover clinical pathways and evidence-based treatment protocols.

Predictive Maintenance	Mine machine sensor event sequences to predict failure patterns before equipment breaks down.
------------------------	---

MODULE 6: Text Mining (Lectures 11–12)

Lecture 11: Text Databases

11.1 The Unstructured Data Problem

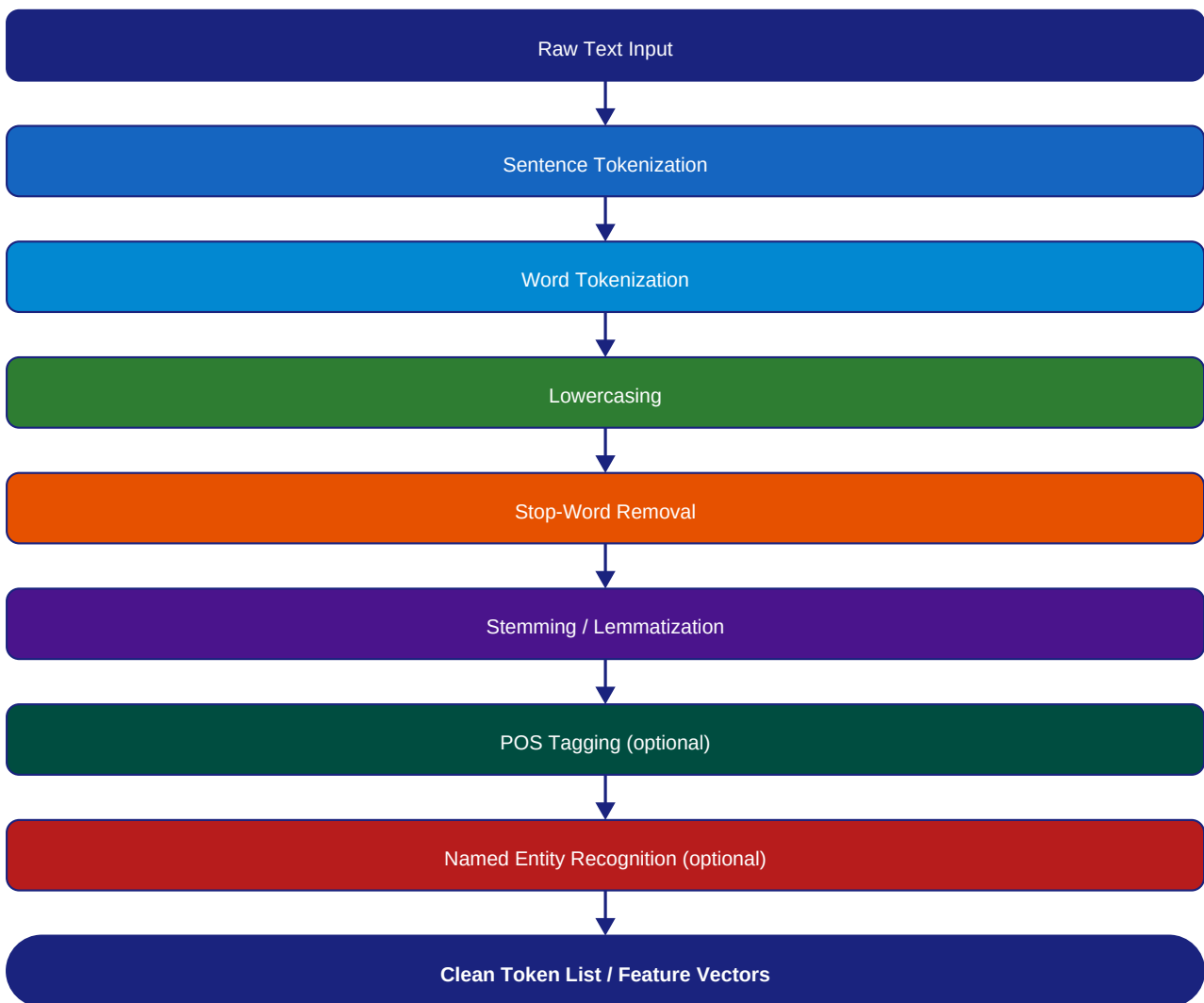
Over 80% of enterprise data is unstructured, with text being the dominant form. Emails, reports, social media posts, legal documents, medical records, and web pages all contain valuable knowledge that is inaccessible to traditional structured query systems. Text mining (also called Text Knowledge Discovery in Databases, KDD) bridges this gap by applying NLP and machine learning to extract structured information from text.

Text Mining is the process of deriving high-quality information from text. It involves structuring input text, finding patterns, and finally evaluating and interpreting the output. It combines NLP, IR (Information Retrieval), machine learning, and statistics.

11.2 Text Preprocessing Pipeline

Raw text cannot be directly mined. A sequence of preprocessing steps transforms raw text into a clean, structured representation:

Text Preprocessing Pipeline



Step	Description	Example
Tokenization	Splitting text into individual tokens (words, punctuation, numbers). Sentence tokenization first splits text into sentences. Word tokenization then splits sentences into words. Handles edge cases like contractions (can't → can, not), hyphenated words, URLs.	'Data mining is fun!' → ['Data', 'mining', 'is', 'fun']
Lowercasing	Convert all tokens to lowercase to reduce vocabulary size. Ensures 'Mining' and 'mining' are treated as the same word. Be careful with proper nouns (US vs. us).	'Data Mining' → 'data mining'

Stop-Word Removal	Eliminate high-frequency, semantically weak words (a, an, the, is, are, was) that don't contribute to document meaning. Language-specific stop-word lists are used. Reduces feature space by 30–50%.	['the', 'cat', 'sat', 'on', 'mat'] → ['cat', 'sat', 'mat']
Stemming	Reduces words to their root form by stripping suffixes using heuristic rules. Fast but may produce non-words. Porter Stemmer: 'running' → 'run', 'studies' → 'studi'. Snowball Stemmer handles multiple languages.	'running', 'runs', 'ran' → 'run'
Lemmatization	Converts words to their dictionary base form (lemma) using morphological analysis and POS tags. Slower than stemming but produces valid words. SpaCy, NLTK WordNetLemmatizer. Preferred over stemming for NLP tasks.	'better' → 'good', 'ran' → 'run', 'studies' → 'study'
POS Tagging	Assigns grammatical categories (noun, verb, adjective, adverb) to each token. Enables filtering (keep only nouns+verbs) and improves lemmatization accuracy. HMM, CRF, or Transformer-based taggers.	'Apple/NNP is/VBZ looking/VBG at/IN buying/VBG'
Named Entity Recognition	Identifies and classifies named entities: persons (PER), organizations (ORG), locations (LOC), dates (DATE), monetary values (MONEY). Enables information extraction from text. spaCy, Stanford NER, BERT-NER.	'Barack Obama visited New Delhi in January 2024'

Lecture 12: Text Mining Techniques

12.1 Text Representation Models

After preprocessing, text must be converted to numerical representations. The main models are:

Representation	Description
Bag of Words (BoW)	Represent each document as a vector of word frequencies. Vocabulary size = vector dimension. Ignores word order and grammar. Simple but effective baseline. Sparse vectors.

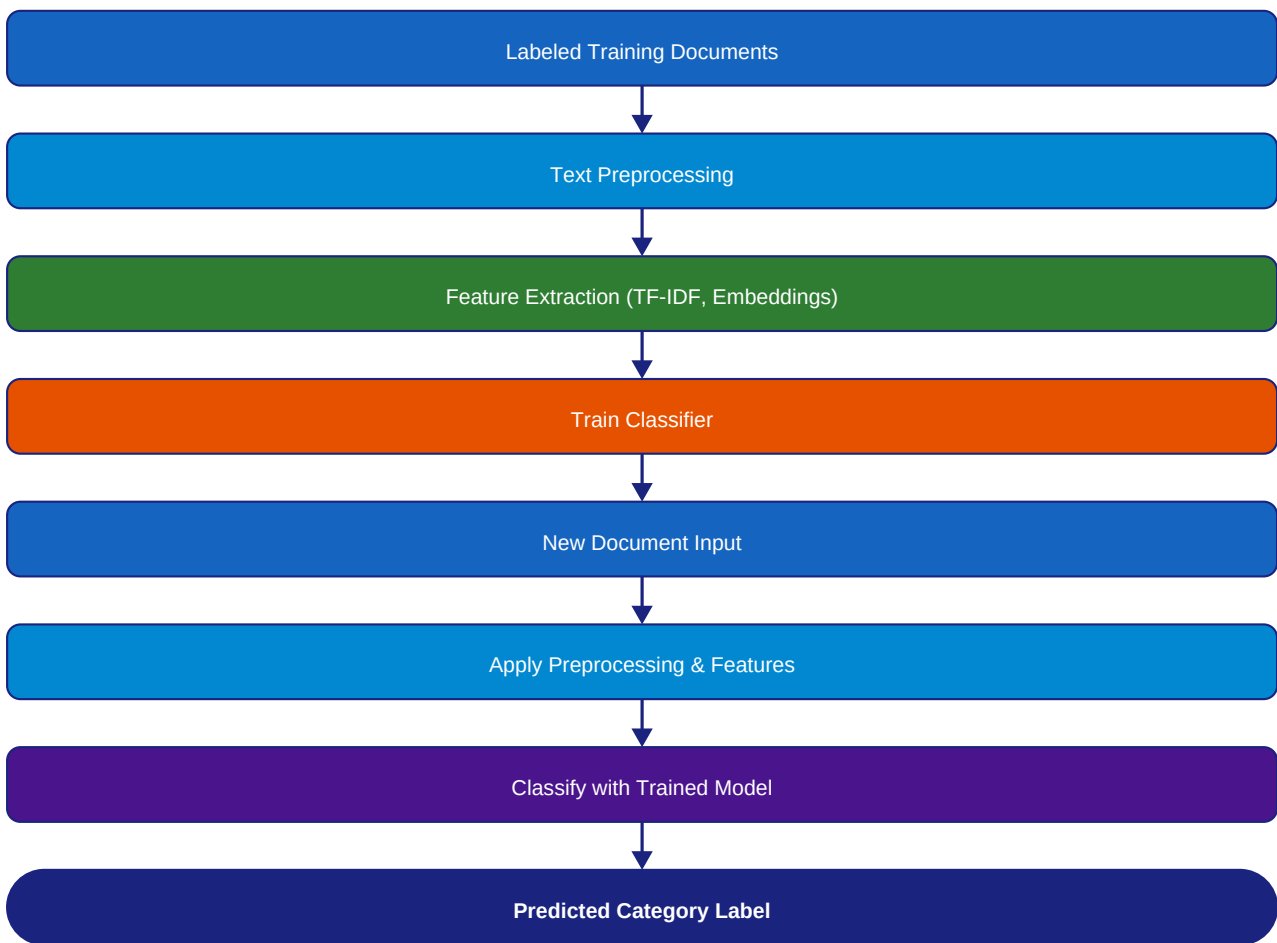
TF-IDF (Term Frequency – Inverse Document Frequency)	Weights terms by their frequency in the document (TF) and rarity across the corpus (IDF). Reduces the weight of common words, highlights discriminative terms. $TF(t,d) = \text{count}(t)/ d $; $IDF(t) = \log(N/df(t))$; $TF-IDF = TF \times IDF$
N-grams	Extend BoW to sequences of n consecutive words. Bigrams (n=2): 'data mining', 'machine learning'. Captures some local word order and phrases. Vocabulary grows rapidly with n.
Word Embeddings (Word2Vec, GloVe, FastText)	Dense low-dimensional vectors (100–300 dim) that capture semantic relationships. Similar words have similar vectors: king - man + woman $\approx$ queen. Trained on large corpora. Transfer learning for downstream tasks.
Contextual Embeddings (BERT, GPT)	Transformer-based models that generate context-dependent word representations. Same word has different vector in different contexts. State-of-the-art for all NLP tasks. Fine-tuned for specific tasks.

$$TF-IDF(t, d) = [\text{count}(t,d) / \sum \text{count}(t', d)] \times \log(N / |\{d: t \in d\}|)$$

12.2 Document Classification

Document classification assigns predefined category labels to text documents. It is a supervised learning task requiring a labeled training corpus.

Document Classification Pipeline



Classifier	Description for Text Classification
Naïve Bayes	Probabilistic classifier based on Bayes' theorem with word independence assumption. Very fast, works well with small data. Strong baseline for text.
SVM (Linear)	Finds maximum-margin hyperplane in feature space. Excellent for text — high-dimensional, sparse features. RBF kernel for non-linear boundaries.
Logistic Regression	Linear model with sigmoid output. Interpretable coefficients show word importance. Regularization (L1/L2) prevents overfitting.
Random Forest	Ensemble of decision trees. Handles non-linear patterns. Feature importance for interpretability. Slower than linear methods for text.
Transformer (BERT)	Fine-tuned pre-trained transformer. State-of-the-art accuracy. Requires GPU and large datasets. Understands context and semantics deeply.

12.3 Sentiment Analysis

Sentiment analysis (opinion mining) determines the emotional tone (positive, negative, neutral) expressed in text. It is one of the most widely applied text mining techniques.

Analysis Level	Description
Document-level	Assign overall sentiment to the entire document. Binary (pos/neg) or multi-class (1–5 stars).
Sentence-level	Determine sentiment of individual sentences. Handles mixed-sentiment documents.
Aspect-level	Identify sentiment towards specific aspects/entities. 'The food was great but the service was slow.' → food: positive, service: negative.
Comparative Sentiment	Detect comparisons between entities: 'Phone A has better battery than Phone B.'

Approaches: (1) Lexicon-based: Use sentiment dictionaries (SentiWordNet, AFINN, VADER) to score words. Fast and interpretable, no training needed. (2) ML-based: Train classifier on labeled data. Higher accuracy but needs labeled training set. (3) Deep learning: LSTM, Transformer — handles negation, sarcasm, and context better than lexicon methods.

12.4 Information Extraction

Information Extraction (IE) automatically extracts structured information (facts, relationships) from unstructured text. Key IE tasks:

- Named Entity Recognition (NER): Extract person names, organizations, locations, dates
- Relation Extraction: Identify relationships between entities ('Apple acquired Shazam')
- Event Extraction: Detect events and their participants ('Earthquake struck Japan on Jan 1')
- Coreference Resolution: Identify when different expressions refer to the same entity ('Steve Jobs... he... Apple's founder...')

12.5 Applications of Text Mining

Text Mining Application Domains



Search Engines: Google, Bing use TF-IDF variants combined with PageRank and neural ranking (BERT-based) to retrieve and rank web pages. Query understanding, spell correction, and query expansion are all text mining applications.

Medical Text Mining: Extracting drug-disease relationships, adverse events, and clinical trial results from PubMed abstracts. Pharmaco-vigilance uses text mining on social media to detect unreported drug side effects.

MODULE 7: Applications of Data Mining (Lectures 13–14)

Lecture 13: Real-World Applications

13.1 E-Commerce: Recommendation Systems

Recommendation systems analyze user behavior, preferences, and item attributes to suggest relevant products, content, or services. They are arguably the most economically impactful data mining application, driving 35% of Amazon's revenue and 80% of Netflix viewing time.

Collaborative Filtering

Recommends items based on preferences of similar users. **User-based CF**: 'Users like you also liked X.' **Item-based CF**: 'People who bought A also bought B.' Uses cosine similarity or Pearson correlation on user-item matrices. Matrix factorization (SVD, ALS) handles scalability.

Content-Based Filtering

Recommends items similar to those the user previously liked, based on item features. Movie recommendation: if user liked action films with specific actors, recommend similar films. Uses TF-IDF for text content, feature vectors for multimedia.

Hybrid Methods

Combine collaborative and content-based filtering to overcome their individual limitations (cold-start problem, sparsity). Netflix Prize winner used blended ensemble of 100+ recommendation models.

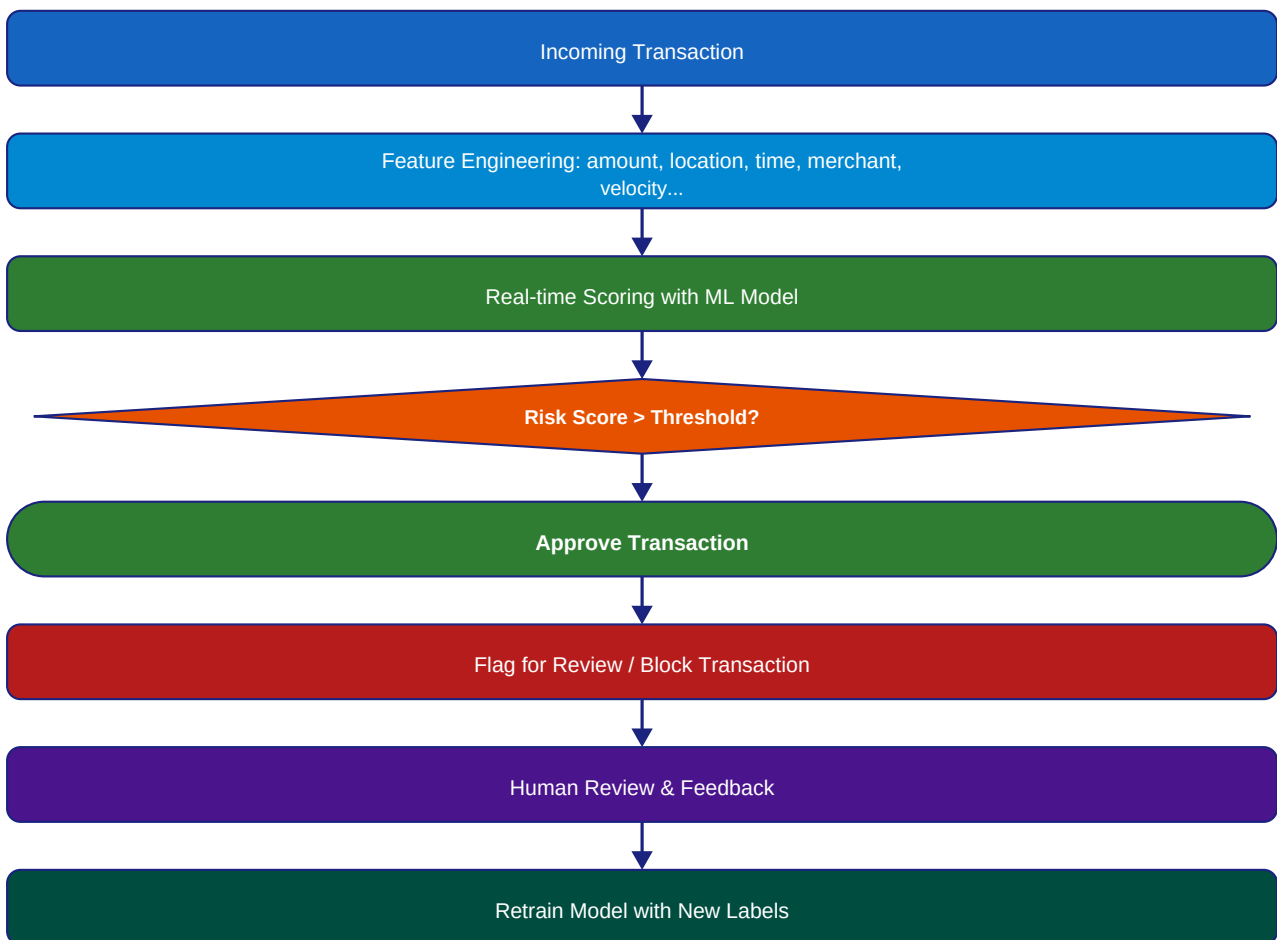
Knowledge-Based

Use domain knowledge and explicit user requirements when preference history is insufficient. Case-based reasoning to find items satisfying stated constraints.

13.2 Banking: Fraud Detection

Banking fraud costs institutions hundreds of billions annually. Data mining enables real-time fraud detection by identifying anomalous transaction patterns.

Real-Time Fraud Detection Pipeline



Key Techniques: Logistic regression (interpretable, regulatory compliant), Random Forest and Gradient Boosting (XGBoost) for high accuracy, Autoencoders for unsupervised anomaly detection, Graph-based methods to detect fraud rings by analyzing transaction networks. Critical challenge: highly imbalanced classes (fraud <0.1% of transactions) — requires SMOTE, cost-sensitive learning, or calibrated thresholds.

13.3 Healthcare: Disease Prediction & Clinical Mining

Healthcare data mining transforms patient records, medical images, genomic data, and clinical trial results into actionable medical knowledge.

Application	Description
Disease Prediction	Train classifiers on patient demographics, lab results, and medical history to predict risk of diabetes, heart disease, cancer. Early intervention improves outcomes.
Clinical Decision Support	Rule-based and ML systems suggest diagnoses, treatment options, and drug dosages based on patient data and medical knowledge bases.
Survival Analysis	Kaplan-Meier curves and Cox proportional hazards models analyze time-to-event data (patient survival, disease recurrence).

Drug Discovery	Mine biomedical literature and molecular databases to identify drug candidates, repurpose existing drugs, and predict drug-drug interactions.
Hospital Operations	Predict patient admission rates, ICU occupancy, and readmission risks to optimize resource allocation and staffing.
Genomics	Mine genomic sequences to identify disease-associated variants (GWAS), discover biomarkers, and enable personalized medicine.

13.4 Social Media Analytics

Social media platforms generate massive streams of user-generated content. Mining this data reveals public opinion, trends, and influence patterns.

Application	Description
Trend Detection	Identify trending topics using frequency analysis and burst detection on hashtags, keywords over time windows.
Community Detection	Graph mining on social networks to identify communities, influencers, and information propagation paths. Algorithms: Louvain, Girvan-Newman.
Bot Detection	Classify accounts as human or automated (bots) based on behavioral patterns: posting frequency, follower ratios, content diversity.
Viral Content Prediction	Predict which posts/videos will go viral based on early engagement signals, network position, and content features.
Crisis Monitoring	Real-time sentiment and topic monitoring during disasters, elections, or product launches for situational awareness.

Lecture 14: System Products & Research Prototypes

14.1 Data Mining Tools

Tool	Description	Best For
WEKA (Waikato Environment for Knowledge Analysis)	Open-source Java-based ML/DM tool from University of Waikato. GUI interface for preprocessing, classification, clustering, association rules, feature selection. Implements 100+ algorithms. Ideal for education and research. Explorer, Experimenter, and KnowledgeFlow interfaces.	Academic research, teaching, rapid prototyping
RapidMiner	Visual workflow-based data mining platform. Drag-and-drop operators. Supports entire DM pipeline: data loading, cleaning, transformation, modeling, evaluation. Large operator library. Community version is free; Enterprise for production.	Business analytics, citizen data scientists

Orange	Python-based open-source data mining suite with visual programming. Widgets for data visualization, preprocessing, modeling. Strong bioinformatics and text mining support.	Research, education, exploratory analysis
Knime	Open-source analytics platform with modular data pipelining. Integration with Python, R, Spark, and cloud services. Thousands of community extensions.	Enterprise analytics, ETL pipelines
Python Ecosystem	scikit-learn (ML algorithms), pandas (data manipulation), NumPy (numerical computing), matplotlib/seaborn (visualization), NLTK/spaCy (NLP), TensorFlow/PyTorch (deep learning).	Industry, research — most widely used
R Ecosystem	Statistical computing language with packages: caret (ML), tidymodels, ggplot2 (visualization), tm (text mining), sp/sf (spatial analysis).	Statistical analysis, academic research

14.2 Research Prototype Systems

Several pioneering research prototype systems have shaped the field of data mining:

System	Contribution
Quest (IBM Almaden)	One of the first data mining research systems. Developed Apriori algorithm for association rule mining. Foundation for many commercial DM systems.
DBMiner	Interactive data mining system tightly integrated with DBMS. Supports OLAP, association rules, classification, clustering on SQL data.
Palau	Parallel association rule mining system. Demonstrated scalability of DM algorithms on distributed databases.
MLC++ / Weka Precursor	Machine learning library systems that led to modern open-source DM toolkits.
SpatialMiner	Research prototype for spatial data mining, integrating GIS with clustering and association rule algorithms.
LexiMine	Text mining research system integrating NLP with data mining for knowledge extraction from large document collections.

14.3 Integration with Databases

Modern data mining is tightly integrated with database systems through:

- **SQL Extensions for Mining:** SQL/DM standard proposes extensions like SELECT MINE to invoke mining algorithms directly within SQL queries.
- **Stored Procedures:** Mine data within the database engine without data movement. Oracle Data Mining, Microsoft SQL Server Analysis Services (SSAS).
- **In-Database Analytics:** Execute R and Python scripts inside the database engine (SQL Server R Services, PostgreSQL PL/Python).
- **Data Warehouse Integration:** Mining models trained on data warehouse contents; results stored back as new aggregations or predictions.

- **PMML (Predictive Model Markup Language):** XML standard for representing mining models, enabling model exchange between different tools.

MODULE 8: Advanced Themes & Trends (Lectures 15–16)

Lecture 15: Additional Themes in Data Mining

15.1 Web Mining

Web mining applies data mining techniques to the World Wide Web to discover patterns in web content, structure, and usage. The web's scale (billions of pages), dynamic nature, and heterogeneity make it both a rich source and a challenging target for knowledge discovery.

Web Content Mining

Mining the actual content (text, images, structured data) of web pages. Includes: web scraping (BeautifulSoup, Scrapy), information extraction from HTML, focused crawling for domain-specific content, and wrapper learning (extracting structured data from templated pages like product listings).

Applications: Price comparison engines, news aggregators, knowledge base construction.

Web Structure Mining

Analyzes the hyperlink structure of the web to infer authority, relevance, and community structure. Key algorithms: **PageRank**: Google's algorithm — iteratively distributes rank through links; pages with many inbound links from high-rank pages get high rank. **HITS (Hyperlink-Induced Topic Search)**: Identifies hub pages (link to authorities) and authority pages (linked from hubs) for a query topic.

Web Usage Mining

Mines web server logs and user click-stream data to understand user navigation patterns. Preprocessing: data cleaning (remove spiders), user session identification, path completion. Mining: sequential pattern mining on click-streams, clustering users by navigation behavior, association rules on page co-visitation. Applications: Website restructuring, personalization, targeted advertising.

15.2 PageRank Algorithm

$$PR(A) = (1 - d) + d \times \sum [PR(T) / C(T)]$$

Where d = damping factor (≈ 0.85), T = pages linking to A , $C(T)$ = outlinks of T . Simulates random surfer model: probability of reaching page A by random walking.

15.3 Graph Mining

Graph mining discovers patterns in graph-structured data. Graphs are ubiquitous: social networks, protein interaction networks, knowledge graphs, road networks, chemical compound structures.

Graph Mining Task	Description
Frequent Subgraph Mining	Find subgraphs appearing in many graphs. Algorithms: gSpan (graph-based Substructure pattern mining), FFSM, Gaston. Used in bioinformatics for drug discovery.
Graph Classification	Classify entire graphs into categories. Feature extraction: graph kernels (Weisfeiler-Lehman), or graph neural networks (GCN, GraphSAGE).

Link Prediction	Predict missing or future edges in a graph. Social networks: friend recommendation. Citation networks: paper references. Drug-target interaction prediction.
Community Detection	Partition graph nodes into cohesive groups (communities). Algorithms: Louvain (modularity optimization), Label Propagation, Girvan-Newman (betweenness-based edge removal).
Node Classification	Predict labels of unlabeled nodes using their features and graph structure. Semi-supervised learning with GNN (Graph Convolutional Networks).
Graph Anomaly Detection	Identify unusual nodes, edges, or subgraphs. Financial fraud ring detection, network intrusion detection.

15.4 Stream Data Mining

Data streams are continuous, potentially infinite sequences of data arriving at high velocity. Traditional algorithms that require multiple passes over the data or unbounded memory cannot be directly applied. Stream mining algorithms must process each item at most once, using limited memory, and produce approximate results.

- **Concept Drift:** The underlying data distribution changes over time (e.g., user preferences evolve). Mining models must adapt continuously.
- **Volume & Velocity:** Millions of events per second (Twitter, IoT sensors, network packets). Algorithms must run in $O(1)$ time per item.
- **Limited Memory:** Cannot store all historical data. Must maintain compact summaries (sketches, synopses, sliding windows).
- **One-Pass Constraint:** Each data item can be processed at most once. No rewind possible.

Algorithm	Description
VFDT (Very Fast Decision Tree)	Hoeffding Tree algorithm for stream classification. Uses Hoeffding bound to decide when enough samples seen to make reliable split.
CluStream	Micro-cluster based stream clustering with offline macro-clustering. Maintains summary statistics in micro-clusters.
Lossy Counting	Approximate frequent itemset mining in streams with guaranteed error bound. Memory $O(1/\epsilon \log(\epsilon n))$.
Reservoir Sampling	Maintain uniform random sample of stream of unknown size. Each item has equal probability of being in reservoir.
Sliding Window	Maintain statistics over the most recent W items. Aging: exponentially decay old observations. ADWIN detects concept drift.

Lecture 16: Trends in Data Mining

16.1 Big Data Mining

Big Data is characterized by the 5 V's: Volume, Velocity, Variety, Veracity, and Value. Traditional single-machine mining algorithms cannot scale to petabyte datasets. Distributed computing frameworks enable large-scale mining.

Big Data Processing Frameworks



Pattern/Framework	Description
MapReduce Pattern	Map: Apply function to each data item independently (embarrassingly parallel). Shuffle: Group intermediate results by key. Reduce: Aggregate values for each key. Example: Word count, k-means iteration, matrix multiplication. All classic DM algorithms can be expressed as MapReduce chains.
Spark MLlib	In-memory distributed ML library. 10-100x faster than MapReduce Hadoop for iterative algorithms (gradient descent, EM). Supports: classification, regression, clustering, collaborative filtering, FP-Growth, PCA, SVD at scale.
Lambda Architecture	Combines batch processing (historical, accurate) with stream processing (real-time, approximate). Batch layer: recomputes views on all historical data. Speed layer: handles recent data with low latency. Serving layer: merges batch and speed layer outputs.

16.2 AI + Data Mining Integration

The boundary between classical data mining and AI/machine learning has largely disappeared. Modern DM is synonymous with applied ML, with deep learning transforming every domain:

AI-DM Integration	Description
Deep Learning for Feature Learning	Convolutional Neural Networks (CNNs) replace hand-crafted image features. Recurrent Networks (LSTM, GRU) and Transformers replace hand-crafted time series features. BERT replaces TF-IDF for text. Deep learning automates the feature extraction step.
Transfer Learning	Pre-trained models (ImageNet CNNs, BERT, GPT) fine-tuned on domain-specific data. Drastically reduces labeled data requirements. Democratizes DM for small organizations.
AutoML (Automated Machine Learning)	Automates algorithm selection, hyperparameter tuning, and feature engineering. Systems: Google AutoML, H2O AutoML, Auto-WEKA, TPOT. Enables non-experts to apply DM effectively.

Explainable AI (XAI)	Regulatory (GDPR) and ethical requirements demand interpretable models. LIME: local surrogate models explain individual predictions. SHAP: Shapley values quantify each feature's contribution. Attention visualization for transformer models.
Federated Learning	Train ML models across distributed datasets without centralizing data. Privacy-preserving: raw data never leaves the device/institution. Critical for healthcare (HIPAA), finance (GDPR), and mobile (on-device learning).

16.3 Spatial Mining Trends

Spatial mining continues to evolve with new data sources and applications:

- **Urban Computing:** Mining heterogeneous urban data (traffic, GPS traces, social check-ins, weather) to understand city dynamics. Microsoft Research Urban Computing group pioneered this field.
- **Trajectory Mining:** Discover patterns in moving object trajectories (taxi GPS data, animal tracking, hurricane paths). Identifies frequent routes, hotspots, and anomalous movements.
- **Remote Sensing + DL:** Deep learning on satellite and aerial imagery for land use classification, crop yield prediction, disaster damage assessment, and infrastructure monitoring at scale.
- **Location-Based Social Networks (LBSN):** Mining Foursquare, Instagram, Twitter check-in data for POI recommendation, mobility pattern discovery, and socio-spatial analysis.
- **Spatial Graph Neural Networks:** Extend GNNs to incorporate spatial coordinates and geographic relationships in node and edge features.

16.4 Web Mining Trends

- **Knowledge Graphs:** Large-scale structured knowledge bases (Wikidata, Google Knowledge Graph, Freebase) built by mining the web. Enable semantic search and question answering.
- **Deep Web Mining:** Querying and extracting data from databases behind web forms (the 'invisible web'). Estimated 500× larger than the surface web.
- **Social Media Mining:** Real-time analytics on Twitter, Reddit, TikTok for market intelligence, political analysis, and public health monitoring.
- **Semantic Web Mining:** Exploiting RDF/OWL structured data (linked data, DBpedia) for richer mining. Combines symbolic AI with statistical ML.

16.5 Privacy & Ethical Concerns

As data mining becomes pervasive, privacy and ethical concerns demand urgent attention:

Privacy-Preserving Data Mining (PPDM)

Techniques that enable mining without revealing sensitive individual information: **Data Perturbation:** Add calibrated random noise to data before mining. **k-Anonymity:** Ensure each record is indistinguishable from k-1 others on quasi-identifiers. **Differential Privacy:** Mathematical guarantee (ϵ -differential privacy) that output doesn't depend significantly on any single individual's data. **Secure Multi-Party Computation:** Multiple parties jointly compute a mining function without revealing their private inputs.

Algorithmic Bias & Fairness

Mining models can perpetuate or amplify societal biases present in training data. Biased hiring algorithms, discriminatory credit scoring, racially biased criminal justice models have been documented. Fairness metrics: demographic parity, equalized odds, individual fairness. Fairness-aware learning algorithms incorporate fairness constraints during training.

Regulatory Compliance

GDPR (EU): Right to explanation for automated decisions, data minimization, consent requirements, right to erasure. **CCPA (California):** Consumer right to know, opt-out of data sale. **HIPAA (US Healthcare):** Protected Health Information (PHI) handling. Data miners must design systems that comply with applicable regulations.

COMPREHENSIVE SUMMARY & KEY FORMULAS

Master Summary: All Modules at a Glance

#	Module	Key Topics
Module 1	Complex Data Types & OLAP	Spatial, Multimedia, Time-Series, Text; Roll-up, Drill-down, Slice, Dice
Module 2	Descriptive Mining	Characterization, Discrimination, AOI generalization, similarity measures
Module 3	Spatial Mining	R-tree indexing, DBSCAN clustering, spatial association rules, GIS applications
Module 4	Multimedia Mining	Color/Texture/Shape features, CBIR, LSH, face recognition, video surveillance
Module 5	Time Series & Sequences	DTW similarity, trend/seasonality, GSP/PrefixSpan algorithms, stock/DNA mining
Module 6	Text Mining	TF-IDF, preprocessing pipeline, document classification, sentiment analysis, NER
Module 7	Applications & Tools	Recommendation (CF/CBF), fraud detection, healthcare prediction, WEKA, RapidMiner
Module 8	Advanced Trends	Web mining (PageRank, HITS), graph mining, stream mining, Big Data, privacy (DP)

Key Formulas Quick Reference

TF-IDF	$TF(t,d) \times \log(N / df(t))$
Euclidean Distance	$d(x,y) = \sqrt{(\sum (x_i - y_i)^2)}$
Cosine Similarity	$\cos(\theta) = (x \cdot y) / (\ x\ \times \ y\)$
Jaccard Similarity	$J(A,B) = A \cap B / A \cup B $
Haversine (Spatial)	$d = 2R \cdot \arcsin(\sqrt{[\sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2)]})$
PageRank	$PR(A) = (1-d) + d \times \sum [PR(T) / C(T)]$
Information Gain	$IG(T, a) = H(T) - \sum [T_v/ T \times H(T_v)]$
Differential Privacy	$P[M(D) \in S] \leq e^{\epsilon} \times P[M(D') \in S] \text{ for neighboring } D, D'$

These notes cover all 8 modules and 16 lectures of the Complex Data Mining course. For exam preparation, focus on: (1) understanding the differences between data types and their appropriate mining techniques, (2) being able to trace algorithm flowcharts step by step, (3) knowing key formulas and when to apply each similarity measure, and (4) connecting theoretical concepts to real-world applications.